ANSWERS

1.

(a) e.g. more (output) power available
e.g. less ripple for same smoothing capacitor
any sensible suggestion B1

(b) **(i)** curve showing half-wave rectification B1

(ii) similar to **(i)** but phase shift of 180° B1

(c) **(i)** correct symbol, connected in parallel with R B1

(ii) **1** larger capacitor / second capacitor in parallel with R B1
(not increase *R*)
2 same peak values B1
correct shape giving less ripple B1

2.

(a) *either* the value of steady / constant voltage — M1
that produces same power (in a resistor) as the alternating voltage — A1
or if alternating voltage is squared and averaged — (M1)
the r.m.s. value is the square root of this averaged value — (A1)

(b) **(i)** 220 V — A1

(ii) 156 V — A1

(iii) 60 Hz — A1

(c) power = V_{rms}^2 / R — C1
$R = 156^2 / 1500$
$= 16\ \Omega$ — A1

3.

(a) power / heating depends on I^2 — M1
so independent of current direction — A1

(b) *either* maximum power = I_0^2R *or* average power = I_{RMS}^2R — M1
$I_0 = \sqrt{2} \times I_{RMS}$ — M1
maximum power = 2 × average power
ratio = 0.5 — A1

4.

(a) **(i)** $2\pi f = 380$ C1

frequency = 60 Hz A1

(ii) $I_{RMS} \times \sqrt{2} = I_0$ C1

$I_{RMS} = 9.9 / \sqrt{2}$

$= 7.0\,A$ A1

(b) power = I^2R C1

$R = 400 / 7.0^2$

$= 8.2\,\Omega$ A1

5.

(a) **(i)** period = 1/50 C1

t_1 = 0.03 s A1

(ii) peak voltage = 17.0 V A1

(iii) r.m.s. voltage = $17.0/\sqrt{2}$

= 12.0 V A1

(iv) mean voltage = 0 A1

(b) power = V^2/R C1

= $12^2/2.4$

= 60 W A1

6.

(a) supply connected correctly (to left & right) B1

load connected correctly (to top & bottom) B1

(b) e.g. power supplied on every half-cycle

greater <u>average/mean</u> power

(any sensible suggestion, 1 mark) B1

(c) **(i)** reduction in the variation of the output voltage/current B1

(ii) larger capacitance produces more smoothing M1

either product *RC* larger

or for the same load A1

7.

(a) **(i)**	peak voltage = 4.0 V	A1
(ii)	r.m.s. voltage (= $4.0/\sqrt{2}$) = 2.8 V	A1
(iii)	period T = 20 ms	M1
	frequency = $1 / (20 \times 10^{-3})$	M1
	frequency = 50 Hz	A0
(b) **(i)**	change = 4.0 – 2.4 = 1.6 V	A1
(ii)	$\Delta Q = C\Delta V$ or $Q = CV$	C1
	$= 5.0 \times 10^{-6} \times 1.6 = 8.0 \times 10^{-6}$ C	A1
(iii)	discharge time = 7 ms	C1
	current = $(8.0 \times 10^{-6}) / (7.0 \times 10^{-3})$	M1
	$= 1.1(4) \times 10^{-3}$ A	A0
(c)	average p.d. = 3.2 V	C1
	resistance = $3.2 / (1.1 \times 10^{-3})$	
	= 2900 Ω *(allow 2800 Ω)*	A1

8.

(a) **(i)**	connection to 'top' of resistor labelled as positive	B1
(ii)	diode B and diode D	B1
(b) **(i)**	V_P = 4.0 V	C1
	mean power = $V_P^2/2R$	C1
	$= 4^2 / (2 \times 2700)$	
	$= 2.96 \times 10^{-3}$ W	A1
(ii)	capacitor, correct symbol, connected in parallel with R	B1
(c)	graph: half-wave rectification	M1
	same period and same peak value	A1

9.

(a) **(i)** $V_0 (= 14\sqrt{2}) = 19.8\ (20)$ V A1

(ii) $\omega (= 2\pi \times 750) = 4700\ \text{rad s}^{-1}$ A1

(b) large amount of charge required to charge capacitor M1
capacitor would charge and discharge rapidly/in a very short time
or capacitor would charge and discharge 750/1500 times per second M1

$I = Q/t$, so large current A1

10.

(a) period = 15 ms C1

frequency $(= 1 / T) = 67$ Hz A1

(b) zero A1

(c) $I_{\text{r.m.s.}} = I_0 / \sqrt{2}$ C1

$= 0.53$ A A1

(d) energy $= I_{\text{r.m.s.}}^{\ 2} \times R \times t$ or $\frac{1}{2} I_0^{\ 2} \times R \times t$
or
power $= I_{\text{r.m.s.}}^{\ 2} \times R$ and energy = power $\times\ t$ C1

energy $= 0.53^2 \times 450 \times 30 \times 10^{-3}$

$= 3.8$ J A1

11.

(a) point P shown at 'lower end' of load B1

(b) $V_{\text{r.m.s.}} = 6.0 / \sqrt{2} = 4.24$ V C1

$I_{\text{r.m.s.}} = 4.24 / (2.4 \times 10^3)$
$= 1.8 \times 10^{-3}$ A A1

(c)	**(i)** capacitor in parallel with load	B1
	(ii) line from peak to curve at 3.0 V for either half- or full-wave rectified	M1
	correct curvature on line (gradient becoming more shallow)	A1
	line drawn as for full-wave rectified	A1

12.

11(a)(i)	circles drawn only around the top left and bottom right diodes	**B1**
11(a)(ii)	B shown as (+)ve and A shown as (–)ve	**B1**
11(b)(i)	$V_{r.m.s.}$ (= 5.6 / √2) = 4.0 V	**A1**
11(b)(ii)	380 = 2π*f* **or** *f* = 60.5 Hz	**C1**
	number (= 2*f*) = 120	**A1**
11(c)(i)	peak values (all) unchanged	**B1**
	(all) minima shown at 4.0 V	**B1**
	three lines from near peak showing concave curves after leaving dotted line not 'kinked' and not cutting the peak reaching candidate's minimum at the point where the decay meets the next dotted line	**B1**
	three lines drawn along the dotted lines showing rise in voltage from minima back to peak values	**B1**
11(c)(ii)	mean p.d. is higher **or** r.m.s. p.d. is higher **or** capacitor supplies energy to resistor	**M1**
	so (mean) power increases	**A1**

13.

10(a)	V_{MAX} = 15 V	**A1**
10(b)	210 = 2π / *T*	**C1**
	T = 0.0299 s	**C1**
	$(t_2 - t_1)$ = 0.060 s	**A1**

14.

10(a)(i)	lower right **and** upper left diodes circled	**B1**
10(a)(ii)	maximum = 7.0√2 = 9.9 V	**A1**
10(b)(i)	correct symbol for capacitor, shown connected in parallel with R	**B1**
10(b)(ii)	**1.** (ripple) decreases	**B1**
	2. (ripple) increases	**B1**

15.

10(a)	**230** V	A1
10(b)	ω = 100π $T = \frac{2\pi}{\omega} = \frac{2\pi}{100\pi}$	C1
	$= 0.020\,s$	A1
10(c)(i)	half-wave (rectification)	B1
10(c)(ii)	sinusoidal half waves in positive V only or negative V only, peak at 320 V	B1
	line at zero for second half of cycle	B1
	two time periods shown, each of 0.020 s	B1
10(c)(iii)	capacitor added in parallel with resistor	B1

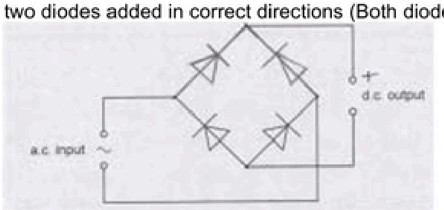
16.

10(a)	the steady current **or** the direct current	M1
	that produces the same heating effect (as the alternating current)	A1
10(b)(i)	peak current = 2.6 A **and** r.m.s. current = 1.8 A	A1
10(b)(ii)	peak current = 2.0 A **and** r.m.s. current = 2.0 A	A1
10(c)(i)	$k = 2\pi f$	C1
	$= 2\pi \times 50$ $= 310$ rad s^{-1}	A1
10(c)(ii)	power = V_{RMS}^2 / R **or** power = $V_0^2 / 2R$	C1
	$R = (240 / \sqrt{2})^2 / 3200$ **or** $R = 240^2 / (2 \times 3200)$ $R = 9.0\,\Omega$	A1

17.

9(a)	constant voltage	M1
	that produces/dissipates same power as (the mean power of) the alternating voltage	A1
9(b)(i)	(maximum) rate of cutting of (magnetic) flux doubles	B1
	(peak and hence) r.m.s. induced e.m.f. doubles	B1
9(b)(ii)	sketch: (sinusoidal) wave of period 10 ms	B1
	peak E shown as ± 34 V *(1 mark out of 2 awarded if peak E shown as* ± 17 V *or* ± 24 V)	B2
9(c)	current in the coil results in forces that oppose its rotation **or** current in the resistor dissipates the energy of rotation	B1
	coil stops rotating	B1

18.

7(a)(i)	two diodes added in correct directions (Both diodes pointing inwards and upwards), correct symbols only 	B1
7(a)(ii)	'+' anywhere on upper output wire	B1
7(b)(i)	$\omega = 2\pi / T$ $= 2\pi / 2.5$ $= 0.80\,\pi$ or $4\pi / 5$ or 2.5	C1
	(V =) **3.5 sin (0.8π *t*)** or 3.5 sin (4π t / 5) or 3.5 sin (2.5 t)	A1
7(b)(ii)	$(P=)\dfrac{V^2}{2R}$ or $(P=)\dfrac{V_{r.m.s.}{}^2}{R}$ $=\dfrac{3.5^2}{2\times 12}$ or $\dfrac{2.47^2}{12}$	C1
	= 0.51 W	A1

19.

5(a)(i)	conversion (from a.c.) to d.c.	B1
5(a)(ii)	full-wave (rectification)	B1
5(b)(i)	P labelled – **and** Q labelled +	B1
5(b)(ii)	V_{OUT} scale labelled 4 and 8 on the 2 cm tick marks	B1
	$T = 2\pi / \omega$ $= 2\pi / 25\pi$ $= 0.08$ s	C1
	t scale labelled 0.02, 0.04, 0.06, 0.08, 0.10, 0.12 on the 2 cm tick marks	A1
5(c)(i)	correct symbol used for capacitor and capacitor connected in parallel with the 1.2 kΩ resistor.	B1
5(c)(ii)	straight lines or curves, with negative decreasing gradients, drawn between adjacent peaks, from top of first peak to meet line going up to next peak	B1
	lines, from one peak to the line going up to the next peak, show a drop in p.d. of 1½ small squares	B1
5(c)(iii)	$V = 0.90 \times 6.0$ (= 5.4 V) **or** discharge time (for each cycle) = 0.034 s	C1
	$V = V_0 \exp(-t / RC)$ $5.4 = 6.0 \exp[-0.034 / (1.2 \times 10^3 \times C)]$	C1
	$C = 2.7 \times 10^{-4}$ **F**	A1

20.

7(a)(i)	peak voltage = $4.2 \times \sqrt{2}$ (= 5.9 V)	B1
	power = V^2 / R = $5.9^2 / 760$ = 0.046 W or 46 mW	A1
7(a)(ii)	sketch shows peak(s) in power at 46 mW	B1
	correct shape (sinusoidal wave sitting on t-axis)	B1
	four cycles of repeating pattern shown, with P = 0 at 0, 10, 20, 30, 40 μs	B1
7(a)(iii)	line is symmetrical about 23 mW	B1
7(b)(i)	(alternating p.d. makes) the crystal vibrate	B1
	vibrations (of crystal) causes air to vibrate	B1
	frequency is in ultrasound range	B1
7(b)(ii)	(air makes) crystal vibrate, which causes an e.m.f. to be generated across the (second) crystal	B1

21.

5(a)(i)	correct circuit symbol for a diode shown correctly connected in series with the wires leading into and out of the dotted box	B1
5(a)(ii)	smoothing / V_{OUT} is smoothed	B1
5(b)(i)	frequency = 1 / 0.04 = 25 Hz	A1
5(b)(ii)	$V = V_0 \exp(-t/RC)$ **and** $\tau = RC$ **or** $V = V_0 \exp(-t/\tau)$	C1
	$3.25 = 5.50 \exp(-0.020/\tau)$ leading to τ = 0.038 s	A1
5(b)(iii)	$\tau = RC$	C1
	capacitance = 0.038 / 14000 = 2.7×10^{-6} F	A1
5(c)	V_{IN} has constant magnitude in both positive and negative directions	B1
	(so) V_{OUT} is (now) constant / V_{OUT} does not vary with time	B1

22.

7(a)(i)	full-wave (rectification)	B1
7(a)(ii)	lower left diode shown pointing left	B1
	lower right and upper left diodes shown pointing left	B1
7(a)(iii)	arrow indicating current direction in resistor to the right	B1
7(b)(i)	sketch: periodic line showing minimum V_{OUT} = 0 and maximum $V_{OUT} = +V_0$	B1
	line showing peak V_{OUT} at t = 0, 0.5T, 1.0T, 1.5T and 2.0T, with V_{OUT} going to zero half-way in between each peak	B1
	line showing correct modulated sine shape	B1
7(b)(ii)	sketch: sinusoidal curve with troughs sitting on the time axis	B1
	<u>peak</u> power at t = 0, 0.5T, 1.0T, 1.5T and 2.0T and zero power half-way in between each peak	B1
7(b)(iii)	same power-time graph with or without rectification, so same V_{rms} **or** V^2-time graph is same for both V_{OUT} and V_{IN}, so same V_{rms} **or** power does not depend on sign of V, so same V_{rms}	B1

23.

7(a)(i)	$P = I_0^2R$	**A1**
7(a)(ii)	$P = 4I_0^2R$	**A1**
7(b)	sketch: square wave of period T, with P always non-zero	**B1**
	horizontal lines, from 0 to $0.5T$ and from $1.0T$ to $1.5T$, all at the same level that the scale indicates to be I_0^2R	**B1**
	horizontal lines, from $0.5T$ to $1.0T$ and from $1.5T$ to $2.0T$, at a level that is four times higher than the lower lines	**B1**
7(c)(i)	$<P> = (5/2)I_0^2R$	**A1**
7(c)(ii)	$<P> = I_{r.m.s.}^2R$	**C1**
	$I_{r.m.s.}^2R = (5/2)I_0^2R$ $I_{r.m.s.} = \sqrt{(5/2)}\ I_0$	**A1**